Cybersecurity Incident Report: DNS Connectivity Issue

Part 1: Incident Summary and tcpdump Analysis

Incident Overview

Users reported that they could not access www.yummyrecipesforme.com and received a “Destination port unreachable” error.

Network traffic was captured and analyzed using tcpdump. The logs showed that DNS queries sent over UDP to port 53 were receiving ICMP error messages.

Key Findings

- **DNS:** Resolves domain names to IP addresses.
- **UDP port 53:** Used to send DNS queries in this scenario.
- **ICMP:** Reports that the destination UDP port is unreachable.
- **Timestamp:** The first recorded event occurred at 13:24:32.192571.

The user's device (192.51.100.15) sent DNS queries to the DNS server (203.0.113.2) to resolve the domain. The same ICMP error occurred multiple times, preventing successful DNS resolution.

Part 2: Analysis and Probable Root Cause

Analysis

The repeated ICMP errors indicate that DNS queries could not reach the destination service through UDP port 53. As a result, the browser could not resolve the domain and proceed normally to the website.

The available evidence identifies a DNS connectivity issue but does not confirm the exact root cause.

Probable Cause

The DNS service may be unavailable or not listening on UDP port 53. A firewall rule or network configuration issue could also be blocking DNS traffic.

Recommended Actions

1. Verify that the DNS service is running and listening on UDP port 53.
2. Review firewall rules and network configurations.
3. Check DNS server logs for errors.
4. Restore the service or correct the configuration as needed.
5. Test DNS resolution and verify HTTPS connectivity afterward.

Conclusion

The investigation identified a DNS resolution failure associated with ICMP “port unreachable” errors. Further investigation is required to confirm the root cause and restore normal access to the website.